



SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

PROF. ARINDAM BANERJEE

Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 04: Statistical Learning Theory-I

Lecture 01: Bias Variance and Mean Squared Errors



What is Statistical Learning Theory?

What is Statistical Learning Theory?

Statistical Learning Theory

- Statistical Learning Theory studies how algorithms can learn relationships between variables from data.
- The goal is to find a rule h_S in order to model Y using a sample data $X_1, X_2, \dots, X_n$.
- It provides a framework for understanding **how well models generalize** to unseen data.

Generalization

- 1 Generalization means performing well on new, unseen data.
- 2 Statistical Learning Theory analyzes why training error and test error differ.

It introduces key ideas such as:

- **Bias–Variance tradeoff**
- **Overfitting and Underfitting**

Main Goal

To understand and analyze how well an algorithm can be generalized.

Key Goals

- To provide a formal mathematical framework for studying how algorithms can infer functions or patterns from limited data samples.
- To understand under what conditions a model trained on finite data can make accurate predictions on new data.
- To Quantify the trade-off between model complexity and training accuracy to avoid overfitting or underfitting.

Error Due to Bias

The error due to bias is the difference between the expected prediction of a model and the correct value of the quantity that is being predicted.

Error Due to Variance

The error due to variance is the variability of a model prediction for a given data point.

High Bias, High Variance

Example

- **High Bias, High Variance:** In this situation, the model fails to capture the true relationship (high bias) and also shows large fluctuations in its predictions across different datasets (high variance). It performs poorly on both training and unseen data.

Suppose you build a model to predict house prices but use a very limited dataset and choose an overly simple model like a linear model with only 2 sample data points. Because the data is small, the model not only misses key patterns (high bias) but also gives very different results when trained on slightly different data samples (high variance).

High Variance, Low Bias

Example

- **High Variance, Low Bias:** In this situation, the model learns the training data too well, so it performs great on training data but poorly on new or unseen data.

Suppose you build a model that uses too many sample points (like wall color, house number etc.). There are now redundant features in the model. It fits the training data almost perfectly (low bias) because it memorizes every small pattern. But when predicting prices for new houses, performance drops sharply because those small details do not generalize.

High Bias, Low Variance

Example

- **High Bias, Low Variance:** In this situation, the model is too simple to capture the true relationship in the data, so it performs poorly both on training and unseen data, but its predictions remain consistent.

Suppose you build a model that predicts house prices using only one feature such as the size of the house. The model ignores other important factors like location, age, and number of rooms. As a result, it systematically underestimates or overestimates prices. Although it gives stable results across datasets (low variance), its predictions are inaccurate overall (high bias).

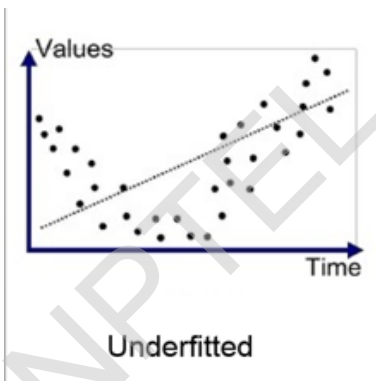
Low Bias, Low Variance

Example

- **Low Bias, Low Variance:** In this situation, the model accurately captures the true relationship in the data and performs well on both training and unseen data. Its predictions are both accurate (low bias) and consistent (low variance).

Suppose you build a model that predicts house prices using multiple relevant features such as size, location, age, and number of rooms. The model is well-tuned and neither too simple nor too complex. As a result, it predicts prices accurately for both training and new houses showing both low bias and low variance.

Under-fitted Model



Using a Straight Line to Fit Curved Data:

- 1 The model is too simple.
- 2 It misses important patterns in the data.
- 3 Predictions are not accurate for training or new data.

Under-fitted Model

Specific Example

Setup:

- 1 An annulus is the region in the plane bounded by two concentric circles:
- 2 We assume that the points are uniformly distributed in this region. That means the joint density in polar coordinates (r, θ) is constant over that area.
- 3 Since $x = r \cos \theta$, $y = r \sin \theta$, and $r = \sqrt{x^2 + y^2}$, the joint probability density function in Cartesian coordinates is

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{\pi(R_{\max}^2 - R_{\min}^2)}, & R_{\min}^2 \leq x^2 + y^2 \leq R_{\max}^2, \\ 0, & \text{otherwise.} \end{cases}$$

Under-fitted Model

Specific Example

Linear Model Fit: Suppose we are trying to fit a linear model for the given data points on the annulus.

- 1 A linear regression model assumes that the data can be approximated by a straight line of the form:

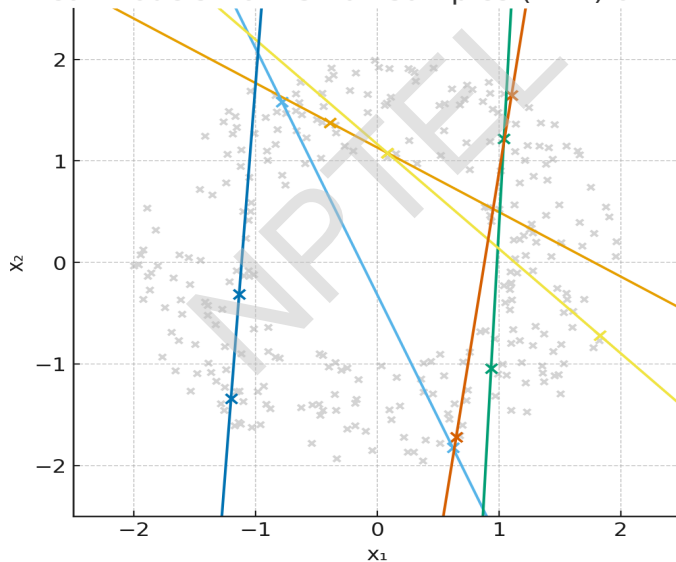
$$Y = \beta_0 + \beta_1 X,$$

- 2 However, in the case of data generated from an annulus, the points do not lie on a straight line but rather form a curve due to their radial symmetry. Specifically, for each θ , the points lie at various radial distances r between $R_{\min}$ and $R_{\max}$, and thus their distribution is circular or ring-shaped, not linear.

Under-fitted Model

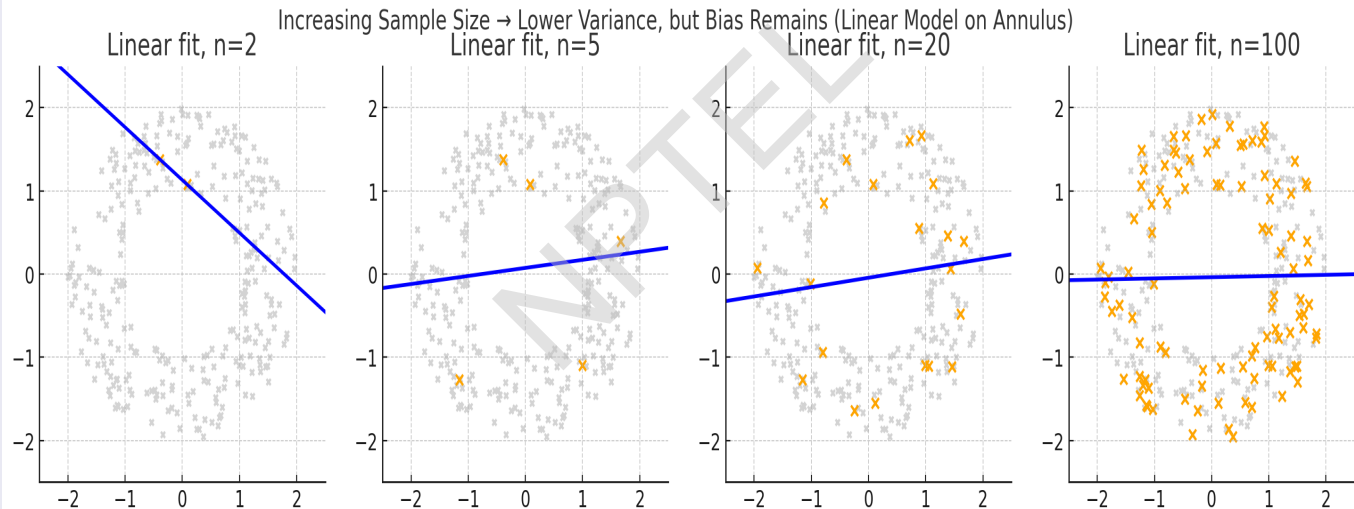
Diagrammatic Representation of Linear Fitting for Sample Size 2:

Biased Linear Models from Small Samples ($n=2$) on Annulus Data



Under-fitted Model

Diagrammatic Representation of Linear Fitting for Different Sample Sizes:



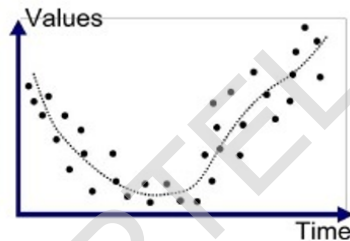
Under-fitted Model

Specific Example

Effect of Sample Size:

- ❶ **Sample Size 2:** In the case of 2 points, the linear regression model will just pass through these two points, but that line will not represent the true underlying relationship. The fit will be biased, as the model assumes a linear relationship when the true relationship is circular.
- ❷ **Higher Sample Size:** As the sample size increases, the linear regression model will be fitted to more points from the annulus.
 - ❶ While the model will still try to minimize the sum of squared residuals, it will continue to assume a linear relationship between X and Y .
 - ❷ However, because the data is inherently radial (non-linear), the resulting regression line will remain biased, even with a large sample size.

Good/Robust-fitted Model

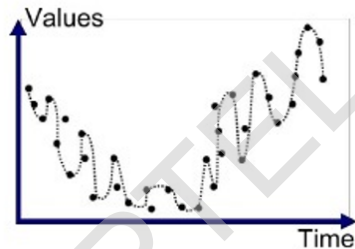


Good Fit/Robust

The Ideal Situation:

- 1 The model captures the main pattern of the data.
- 2 It balances between simplicity and complexity.
- 3 Predictions are accurate and generalize well to new data.

Over-fitted Model



Overfitted

Using a Very Wiggly Curve to Pass Through Every Point:

- 1 The model is too complex.
- 2 It fits the noise in the training data instead of the true pattern.
- 3 Predictions are very accurate on training data but poor on new data.

Over-fitted Model

Specific Example

Setup:

- 1 An annulus is the region in the plane bounded by two concentric circles:
- 2 We assume that the points are uniformly distributed in this region. That means the joint density in polar coordinates (r, θ) is constant over that area.
- 3 Since $x = r \cos \theta$, $y = r \sin \theta$, and $r = \sqrt{x^2 + y^2}$, the joint probability density function in Cartesian coordinates is

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{\pi(R_{\max}^2 - R_{\min}^2)}, & R_{\min}^2 \leq x^2 + y^2 \leq R_{\max}^2, \\ 0, & \text{otherwise.} \end{cases}$$

Over-fitted Model

Specific Example

Fitting a Degree 4 Curve: Suppose we are trying to fit a curve through the given data points

$$(1, 0.01), \quad (0.01, 1), \quad (-1, 0), \quad (0, -1), \quad \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

These five points will uniquely determine a polynomial of degree at most 4 (a quartic polynomial) by interpolation.

Lagrange Interpolation Formula

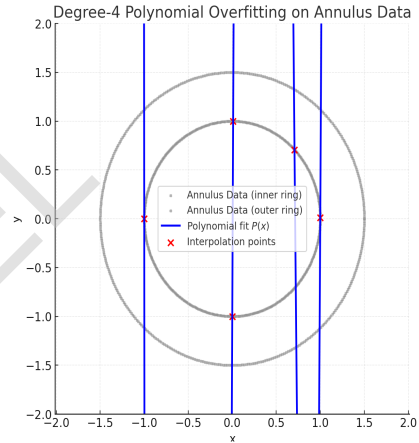
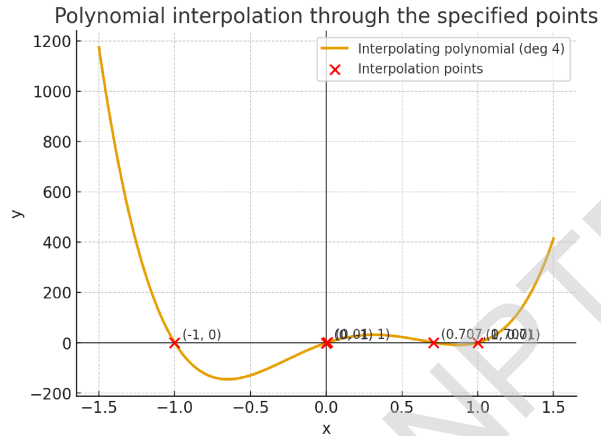
Finding the Degree 4 Curve:

Given data points (x_i, y_i) , the Lagrange interpolating polynomial is

$$P(x) = \sum_{i=1}^n y_i \ell_i(x), \quad \text{where } \ell_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

For our chosen 5 points:

$$P(x) = 282.03x^4 - 202.83x^3 - 281.03x^2 + 202.83x - 1.0.$$



- The polynomial curve bends sharply to touch each point.
- It fits the sample data perfectly but fails to capture the circular annulus trend.
- **High model complexity** can reduce training error to zero but increases generalization error. This illustrates the risk of overfitting.

References

- 1 Shalev-Shwartz, Shai, and Shai Ben-David. Understanding machine learning: From theory to algorithms. Cambridge university press, 2014.
- 2 Lecture 12: Bias-Variance Tradeoff (Machine Learning for Intelligent Systems CS4780/CS5780) by Prof Kilian Weinberger, Cornell University (2018):
<https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote12.html>
- 3 Understanding the Bias-Variance Tradeoff by Scott Fortmann-Roe (2012):
<https://scott.fortmann-roe.com/docs/BiasVariance.html>

Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY
PROF. ARINDAM BANERJEE
Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 04: Statistical Learning Theory-I

Lecture 02: Bias Variance and Mean Squared Errors.



Formal Setup:

- 1 Let $S = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$ be a sample drawn independently and identically from a joint probability distribution $P(X, Y)$.
- 2 We want to find a rule h_S in order to model Y in terms of the sample data S . Note that $y \in \mathbb{R}$.
- 3 We introduce a machine learning algorithm $\mathcal{A}$ on the data set S to learn a hypothesis. Formally, we denote this by $h_S = \mathcal{A}(S)$.

Useful Notations and Terminologies:

- ➊ **Expected value:** For a given $\mathbf{x} \in \mathbb{R}^d$ the expected value $\bar{y}(\mathbf{x})$ is given by:

$$\bar{y}(\mathbf{x}) = E_{y|\mathbf{x}}[Y] = \int_y y \Pr(y|\mathbf{x}) dy$$

- ➋ **Expected Test Error:** For a given h_S learned on the dataset S , the expected test error is:

$$\mathbb{E}_{(\mathbf{x}, y) \sim P(X, Y)} [(h_S(\mathbf{x}) - y)^2] = \int_{\mathbf{x}} \int_y (h_S(\mathbf{x}) - y)^2 \Pr(\mathbf{x}, y) dy d\mathbf{x}$$

- ➌ **Expected Test Error:** For a given algorithm $\mathcal{A}$, the expected test error is:

$$\bar{h} = \mathbb{E}_{\substack{(\mathbf{x}, y) \sim P(X, Y) \\ S \sim P(X, Y)}} [(h_S(\mathbf{x}) - y)^2] = \int_S \int_{\mathbf{x}} \int_y (h_S(\mathbf{x}) - y)^2 \Pr(\mathbf{x}, y) P(S) d\mathbf{x} dy dS$$

where $\Pr(S)$ is the probability of drawing S from a population X, Y with joint distribution $P(X, Y)$

Bias

The error due to bias is the difference between the expected prediction of a model and the correct value of the quantity that is being predicted.

$$\text{Bias}^2 = E_{\mathbf{x}} [(\bar{h}(\mathbf{x}) - \bar{y}(\mathbf{x}))^2]$$

Variance

The error due to variance is the variability of a model prediction for a given data point.

$$\text{Variance} = E_{\mathbf{x}, S} [(h_S(\mathbf{x}) - \bar{h}(\mathbf{x}))^2]$$

Conditional Probability

Conditional Probability

For events A and B with $P(B) > 0$, the **conditional probability** of A given B is

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

Probability of Union (Inclusion-Exclusion)

For any two events A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Example

Suppose a deck of 52 cards:

- Let A = “draw a heart”, B = “draw a queen”
- $P(A) = \frac{13}{52}$, $P(B) = \frac{4}{52}$
- $P(A \cap B) = \frac{1}{52}$ (queen of hearts)
- **Probability of union:**

$$P(A \cup B) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$$

- **Conditional probability:** Probability that it is a queen given it is a heart:

$$P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{1/52}{13/52} = \frac{1}{13}$$

Bayes' Theorem

Setup

Let $\{A_1, A_2, \dots, A_n\}$ be a partition of the sample space Ω , i.e.,

$$A_i \cap A_j = \emptyset \text{ for } i \neq j, \quad \bigcup_{i=1}^n A_i = \Omega.$$

Let B be any event with $P(B) > 0$.

Bayes' Theorem

For any $i = 1, 2, \dots, n$,

$$P(A_i | B) = \frac{P(B | A_i) P(A_i)}{\sum_{j=1}^n P(B | A_j) P(A_j)}$$

where the denominator is the **law of total probability**:

$$P(B) = \sum_{j=1}^n P(B | A_j) P(A_j)$$

Example: Bayes' Theorem

Problem Setup

Suppose there are two bags:

- Bag 1: 3 red balls and 2 blue balls
- Bag 2: 1 red ball and 4 blue balls

A bag is chosen at random with equal probability, and then a ball is drawn.

Question: If a red ball is drawn, what is the probability it came from Bag 1?

Solution Using Bayes' Theorem

Let:

A_1 = "Bag 1 chosen", A_2 = "Bag 2 chosen", B = "Red ball drawn"

$$P(A_1 | B) = \frac{P(B | A_1)P(A_1)}{P(B | A_1)P(A_1) + P(B | A_2)P(A_2)}$$

Compute probabilities:

$$P(A_1) = P(A_2) = 0.5, \quad P(B | A_1) = \frac{3}{5}, \quad P(B | A_2) = \frac{1}{5}$$

$$P(A_1 | B) = \frac{(3/5)(0.5)}{(3/5)(0.5) + (1/5)(0.5)} = \frac{0.3}{0.3 + 0.1} = 0.75$$

Joint Probability Distribution

Let X and Y be two random variables. The **joint probability distribution** describes the probability of all possible pairs (X, Y) .

- **Discrete Case:** The joint probability mass function (pmf) is

$$f_{X,Y}(x, y) = P(X = x, Y = y), \quad \text{for all } x, y.$$

- **Continuous Case:** The joint probability density function (pdf) is

$$f_{X,Y}(x, y) \geq 0, \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1,$$

and for a region $A \subset \mathbb{R}^2$,

$$P((X, Y) \in A) = \int \int_A f_{X,Y}(x, y) dx dy.$$

Example: Joint Distribution (Discrete)

Discrete Case

Let X = number of heads and Y = number of tails when flipping two fair coins. Then the possible pairs (X, Y) and their joint pmf are:

(X, Y)	$(0, 2)$	$(1, 1)$	$(2, 0)$
$f_{X,Y}(x, y)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

That is,

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{4}, & (x, y) = (0, 2) \text{ or } (2, 0), \\ \frac{1}{2}, & (x, y) = (1, 1), \\ 0, & \text{otherwise.} \end{cases}$$

Example: Joint Distribution (Continuous)

Continuous Case

Let (X, Y) be uniformly distributed over the unit square $0 \leq x \leq 1, 0 \leq y \leq 1$.
Then the joint pdf is

$$f_{X,Y}(x, y) = \begin{cases} 1, & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

and it satisfies

$$\int_0^1 \int_0^1 f_{X,Y}(x, y) dx dy = 1.$$

For a region $A \subseteq [0, 1] \times [0, 1]$,

$$P((X, Y) \in A) = \iint_A 1 dx dy = \text{area of } A.$$

Marginal Distributions and Condition for Independence.

- **Marginal Distributions:** The individual distributions can be obtained by summing/integrating over the other variable:

$$f_X(x) = \sum_y f_{X,Y}(x,y) \quad \text{or} \quad f_X(x) = \int_y f_{X,Y}(x,y) dy$$

$$f_Y(y) = \sum_x f_{X,Y}(x,y) \quad \text{or} \quad f_Y(y) = \int_x f_{X,Y}(x,y) dx$$

- **Independence:** X and Y are independent if and only if

$$f_{X,Y}(x,y) = f_X(x)f_Y(y) \quad \text{for all } x,y.$$

Conditional Probability for Random Variables

Discrete Case

Let X and Y be discrete random variables. The **conditional probability mass function** of X given $Y = y$ is

$$P(X = x \mid Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}, \quad P(Y = y) > 0.$$

Continuous Case

Let X and Y be continuous random variables with joint pdf $f_{X,Y}(x, y)$. The **conditional density** of X given $Y = y$ is

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \quad f_Y(y) > 0,$$

where $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$ is the marginal density of Y .

Example: Marginal Distribution (Discrete)

Setup

Roll a fair die once and define:

$$X = \begin{cases} 1, & \text{if the outcome is even,} \\ 0, & \text{if the outcome is odd,} \end{cases} \quad Y = \begin{cases} 1, & \text{if the outcome} > 3, \\ 0, & \text{if the outcome} \leq 3. \end{cases}$$

Marginal Distributions

Outcomes of a fair die: $\{1, 2, 3, 4, 5, 6\}$, each with probability $\frac{1}{6}$.

$$P(X = 1) = P(\text{even}) = \frac{3}{6} = \frac{1}{2}, \quad P(X = 0) = \frac{1}{2},$$

$$P(Y = 1) = P(\text{outcome} > 3) = \frac{3}{6} = \frac{1}{2}, \quad P(Y = 0) = \frac{1}{2}.$$

Example: Conditional Probability (Discrete)

Computing $P(X = 1 \mid Y = 1)$

$$P(X = 1 \mid Y = 1) = \frac{P(X = 1, Y = 1)}{P(Y = 1)}.$$

From the sample space:

$Y = 1$ then the outcome of the die is $\{4, 5, 6\}$, $X = 1$ then the outcome of the die is $\{2, 4, 6\}$.

Thus,

$X = 1, Y = 1$ then the outcome of the dice is the pair $\{4, 6\}$, $P(X = 1, Y = 1) = \frac{2}{6}$, $P(Y = 1) = \frac{3}{6}$.

Hence,

$$P(X = 1 \mid Y = 1) = \frac{\frac{2}{6}}{\frac{3}{6}} = \frac{2}{3}.$$

Example: Marginal Distribution (Continuous)

Setup

Let (X, Y) be continuous random variables with joint pdf

$$f_{X,Y}(x, y) = \begin{cases} 2, & 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Finding the Marginal of Y

To find the marginal pdf of Y :

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx = \int_0^y 2 dx = 2y, \quad 0 < y < 1.$$

Finding the Marginal of X

Similarly,

$$f_X(x) = \int_x^1 2 dy = 2(1 - x), \quad 0 < x < 1.$$

Example: Conditional Density (Continuous)

Conditional Density of X given $Y = y$

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \frac{2}{2y} = \frac{1}{y}, \quad 0 < x < y < 1.$$

Interpretation

Given that $Y = y$, the random variable X is uniformly distributed on $(0, y)$:

$$X | Y = y \sim \text{Uniform}(0, y).$$

This means that once we know $Y = y$, the value of X is equally likely anywhere between 0 and y .

Conditional Expectation

$$E[X | Y = y] = \sum_x x P(X = x | Y = y) \quad (\text{discrete})$$

$$E[X | Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx \quad (\text{continuous})$$

Example

Let X and Y be the outcomes of two fair dice:

- X = outcome of the first die
- Y = outcome of the second die
- Since X and Y are independent:

$$P(X = x | Y = y) = P(X = x) = \frac{1}{6}, \quad x = 1, 2, \dots, 6$$

- Conditional expectation:

$$E[X | Y = y] = \sum_{x=1}^6 x \cdot \frac{1}{6} = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

- **Observation:** $E[X | Y = y] = E[X]$ because X and Y are independent

Properties of Expectation

Let X and Y be two random variables and $a, b, c \in \mathbb{R}$. We have the following properties:

① **Linearity:**

$E[aX + bY] = aE[X] + bE[Y]$ holds even if X and Y are dependent.

② **Expectation of a Constant:**

$$E[c] = c$$

③ **Constant Factor Rule:**

$$E[aX] = aE[X]$$

④ **Additivity:**

$$E[X + Y] = E[X] + E[Y]$$

Properties of Variance

Let X and Y be two random variables and $a, b, c \in \mathbb{R}$. We have the following properties:

❶ **Definition:**

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

❷ **Scaling Property:**

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

❸ **Shift Invariance:**

$$\text{Var}(X + c) = \text{Var}(X)$$

❹ **Variance of a Constant:**

$$\text{Var}(c) = 0$$

❺ **Variance of a Sum of Random Variables (If X and Y are independent :**

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

Expectation of a Sum of Random Variables

Let $X_1, X_2, \dots, X_n$ be random variables and define $S_n = \sum_{i=1}^n X_i$. then, $E[S_n] = \sum_{i=1}^n E[X_i]$.

Proof:

- For simplicity, start with $n = 2$:

$$E[X_1 + X_2] = \sum_{x_1, x_2} (x_1 + x_2) f_{X_1, X_2}(x_1, x_2)$$

for discrete variables, or

$$E[X_1 + X_2] = \int_{\mathbb{R}^2} (x_1 + x_2) f_{X_1, X_2}(x_1, x_2) dx_1 dx_2$$

for continuous variables.

Proof:

- Split the sum/integral using linearity of integration:

$$E[X_1 + X_2] = \sum_{x_1, x_2} x_1 f_{X_1, X_2}(x_1, x_2) + \sum_{x_1, x_2} x_2 f_{X_1, X_2}(x_1, x_2).$$

- In the first term, sum (or integrate) over x_2 :

$$\sum_{x_2} f_{X_1, X_2}(x_1, x_2) = f_{X_1}(x_1),$$

giving

$$\sum_{x_1} x_1 f_{X_1}(x_1) = E[X_1].$$

Similarly, the second term gives $E[X_2]$.

- Therefore,

$$E[X_1 + X_2] = E[X_1] + E[X_2].$$

- The argument extends directly to n variables by induction:

$$E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i].$$

Variance of Sum of Random Variables

Setup

Let $X_1, X_2, \dots, X_n$ be random variables and define $S_n = \sum_{i=1}^n X_i$

- By definition:

$$\text{Var}(S_n) = E[(S_n - E[S_n])^2] = E[S_n^2] - (E[S_n])^2$$

- Expand the square:

$$S_n^2 = \sum_{i=1}^n X_i^2 + 2 \sum_{1 \leq i < j \leq n} X_i X_j$$

- Take expectations:

$$E[S_n^2] = \sum_{i=1}^n E[X_i^2] + 2 \sum_{1 \leq i < j \leq n} E[X_i X_j]$$

$$(E[S_n])^2 = \sum_{i=1}^n (E[X_i])^2 + 2 \sum_{1 \leq i < j \leq n} E[X_i]E[X_j]$$

Variance of Sum of Random Variables

- Subtract to get variance:

$$\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j)$$

- **Special case: Independence:** If X_i are independent,

$$\text{Var}(S_n) = \sum_{i=1}^n \text{Var}(X_i)$$

Law of Iterated Expectation for Random Vectors

Setup

Let

$$\mathbf{X} = (X_1, X_2, \dots, X_n), \quad \mathbf{Y} = (Y_1, Y_2, \dots, Y_n)$$

be random vectors defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Assume that the joint distribution of $(\mathbf{X}, \mathbf{Y})$ admits a density (or mass function) $f_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y})$ such that all expectations below are well-defined.

Statement of the Property

Law of Iterated Expectation (Vector Form)

For any integrable function $g : \mathbb{R}^{2n} \rightarrow \mathbb{R}$,

$$\mathbb{E}_{\mathbf{X}, \mathbf{Y}}[g(\mathbf{X}, \mathbf{Y})] = \mathbb{E}_{\mathbf{X}}[\mathbb{E}_{\mathbf{Y}|\mathbf{X}}[g(\mathbf{X}, \mathbf{Y})]].$$

Interpretation

- The inner expectation averages $g(\mathbf{X}, \mathbf{Y})$ over the conditional distribution of $\mathbf{Y}$ given $\mathbf{X}$.
- The outer expectation then averages over the marginal distribution of $\mathbf{X}$.

Proof Sketch

Proof.

- 1 Write the joint expectation as

$$\mathbb{E}[g(\mathbf{X}, \mathbf{Y})] = \int_{\mathcal{X}} \int_{\mathcal{Y}} g(\mathbf{x}, \mathbf{y}) p_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y}) d\mathbf{y} d\mathbf{x}.$$

- 2 Factorize the joint density:

$$p_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y}) = p_{\mathbf{Y}|\mathbf{X}}(\mathbf{y} | \mathbf{x}) p_{\mathbf{X}}(\mathbf{x}).$$

- 3 Substitute and rearrange:

$$\mathbb{E}[g(\mathbf{X}, \mathbf{Y})] = \int_{\mathcal{X}} \left(\int_{\mathcal{Y}} g(\mathbf{x}, \mathbf{y}) p_{\mathbf{Y}|\mathbf{X}}(\mathbf{y} | \mathbf{x}) d\mathbf{y} \right) p_{\mathbf{X}}(\mathbf{x}) d\mathbf{x}.$$

- 4 Recognize the inner integral as $\mathbb{E}_{\mathbf{Y}|\mathbf{X}=\mathbf{x}}[g(\mathbf{x}, \mathbf{Y})]$ and the outer integral as $\mathbb{E}_{\mathbf{X}}[\cdot]$.

$$\therefore \mathbb{E}[g(\mathbf{X}, \mathbf{Y})] = \mathbb{E}_{\mathbf{X}}[\mathbb{E}_{\mathbf{Y}|\mathbf{X}}[g(\mathbf{X}, \mathbf{Y})]].$$



Sum of Well-Known Discrete Random Variables

Examples

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables:

- **Bernoulli:**

$$X_i \sim \text{Ber}(p), \quad S_n = \sum_{i=1}^n X_i \sim \text{B}(n, p)$$

- **Poisson:**

$$X_i \sim \text{Pois}(\lambda_i), \quad S_n = \sum_{i=1}^n X_i \sim \text{Pois}\left(\sum_{i=1}^n \lambda_i\right)$$

- **Geometric:**

$$X_i \sim \text{Geom}(p), \quad S_n = \sum_{i=1}^n X_i \sim \text{NB}(n, p)$$

- **Binomial:**

$$X_i \sim \text{B}(n_i, p), \quad S_n = \sum_{i=1}^n X_i \sim \text{B}\left(\sum_{i=1}^n n_i, p\right)$$

Sum of Well-Known Continuous Random Variables

Examples

Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables:

- **Normal:**

$$X_i \sim \mathcal{N}(\mu_i, \sigma_i^2), \quad S_n = \sum_{i=1}^n X_i \sim \mathcal{N}\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$$

- **Exponential:**

$$X_i \sim \text{Exp}(\lambda), \quad S_n = \sum_{i=1}^n X_i \sim \text{Gamma}(n, \lambda)$$

- **Chi-square:**

$$X_i \sim \chi^2(k_i), \quad S_n = \sum_{i=1}^n X_i \sim \chi^2\left(\sum_{i=1}^n k_i\right)$$

References

- 1 Shalev-Shwartz, Shai, and Shai Ben-David. Understanding machine learning: From theory to algorithms. Cambridge university press, 2014.
- 2 Lecture 12: Bias-Variance Tradeoff (Machine Learning for Intelligent Systems CS4780/CS5780) by Prof Kilian Weinberger, Cornell University (2018):
<https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote12.html>
- 3 Understanding the Bias-Variance Tradeoff by Scott Fortmann-Roe (2012):
<https://scott.fortmann-roe.com/docs/BiasVariance.html>

Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

PROF. ARINDAM BANERJEE

Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 04: Statistical Learning Theory-I

Lecture 03: Bias Variance and Mean Squared Errors.



Setup

Formal Setup:

Step 1: Drawing a Data Set.

- 1 Let $S = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$ be a sample drawn independently and identically from a joint probability distribution $P(X, Y)$.
- 2 We want to find a rule $Y = h_S(X)$ in order to model Y in terms of the sample data S . Note that $y \in \mathbb{R}$.

Note:

- 1 It may happen that for an input x , there may not be a unique value y .

Example

Let x represent the features of a house and y denote its selling price. It is possible that two houses with identical features may have different selling prices.

Useful Notations and Terminologies:

❶ **Expected value:** For a given $\mathbf{x} \in \mathbb{R}^d$ the expected value $\bar{y}(\mathbf{x})$ is given by:

$$\bar{y}(\mathbf{x}) = E_{y|\mathbf{x}}[Y] = \int_y y \Pr(y|\mathbf{x}) dy$$

In all the expressions of expectation, we loosely use the integration sign. In case of discrete distributions the integration is replaced by summation.

Expected value Example

Example

Let $\mathbf{x} = (3, 1)$. Assume that, X and Y follow a joint distribution such that X has two features and given $\mathbf{x}$, the value Y takes on discrete values according to the following conditional probability distribution:

y	1	2	3
$\Pr(y \mathbf{x})$	0.2	0.5	0.3

The expected value for $\mathbf{x} = (3, 1)$ is computed as:

$$\bar{y}(\mathbf{x}) = \sum_y y \Pr(y|\mathbf{x})$$

$$\begin{aligned}\bar{y}(\mathbf{x}) &= (1)(0.2) + (2)(0.5) + (3)(0.3) \\ &= 0.2 + 1.0 + 0.9 \\ &= 2.1\end{aligned}$$

Interpretation: For the vector $\mathbf{x} = (3, 1)$, the expected (predicted) value of Y is:

$$\bar{y}(\mathbf{x}) = 2.1$$

Setup

Formal Setup:

Step 2: We introduce an algorithm $\mathcal{A}$ on the data set S to learn a hypothesis. Formally, we denote this by $h_S = \mathcal{A}(S)$.

Useful Notations and Terminologies:

- 2) **Expected Test Error of a Given Sample:** For a given h_S learned on the dataset S , the expected test error is:

$$\mathbb{E}_{(\mathbf{x}, y) \sim P(X, Y)} [(h_S(\mathbf{x}) - y)^2] = \int_{\mathbf{x}} \int_y (h_S(\mathbf{x}) - y)^2 \Pr(\mathbf{x}, y) dy d\mathbf{x}$$

Expected Test Error of a given sample Example

Example

Using the same setting and $\mathbf{x} = (3, 1)$:

y	1	2	3
$\Pr(y \mathbf{x})$	0.2	0.5	0.3

Let a simple model be:

$$h_S(\mathbf{x}) = 2$$

Then, the expected test error of a given sample is:

$$\begin{aligned}\mathbb{E}_{y|\mathbf{x}}[(h_S(\mathbf{x}) - y)^2] &= \sum_y (h_S(\mathbf{x}) - y)^2 \Pr(y|\mathbf{x}) \\ &= (2 - 1)^2(0.2) + (2 - 2)^2(0.5) + (2 - 3)^2(0.3) \\ &= 0.2 + 0 + 0.3 = 0.5\end{aligned}$$

Interpretation: For the model $h_S(\mathbf{x}) = 2$, the expected test (mean squared) error is 0.5

Setup

Formal Setup:

Step 3: Realize that h_S is a random variable:

- 1 $S = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$ is an i.i.d. sample that is drawn from a population X, Y with joint distribution $P(X, Y)$ therefore S is a random variable.
- 2 h_S is a function of S and is therefore a random variable.

Useful Notations and Terminologies:

- ④ **Expected Value of Predictor:** For a given algorithm $\mathcal{A}$, the expected classifier is:

$$\bar{h} = \mathbb{E}_{S \sim P(X,Y)}[h_S] = \int_S h_S \Pr(S) dS,$$

where $\Pr(S)$ is the probability of drawing S from a population (X, Y) with joint distribution $P(X, Y)$.

Note:

- ① $\bar{h}$ is the weighted average over functions.

Expected Value of Predictor Example

Example

Assume that depending on the dataset S , the algorithm $\mathcal{A}$ produces different predictors:

S	S_1	S_2	S_3
$h_S(\mathbf{x})$	1.8	2.0	2.4
$\Pr(S)$	0.3	0.4	0.3

Then, the expected value of the predictor is:

$$\bar{h}(\mathbf{x}) = \mathbb{E}_S[h_S(\mathbf{x})] = \sum_S h_S(\mathbf{x}) \Pr(S)$$

$$\begin{aligned}\bar{h}(\mathbf{x}) &= (1.8)(0.3) + (2.0)(0.4) + (2.4)(0.3) \\ &= 0.54 + 0.8 + 0.72 = 2.06\end{aligned}$$

Interpretation: The expected predictor (average model output) is:

$$\bar{h}(\mathbf{x}) = 2.06$$

Useful Notations and Terminologies:

- **Expected Total Test Error:** For a given algorithm $\mathcal{A}$, the expected test error is:

$$\mathbb{E}_{\substack{(\mathbf{x}, y) \sim P(X, Y) \\ S \sim P(X, Y)}} [(h_S(\mathbf{x}) - y)^2] = \int_S \int_x \int_y (h_S(\mathbf{x}) - y)^2 \Pr(\mathbf{x}, y) P(S) \, dx dy dS$$

where $\Pr(S)$ is the probability of drawing S from a population X, Y with joint distribution $P(X, Y)$

Expected Total Test Error Example

Example

Using the same setup:

S	S_1	S_2	S_3
$h_S(\mathbf{x})$	1.8	2.0	2.4
$\Pr(S)$	0.3	0.4	0.3

and for every $\mathbf{x}$

y	1	2	3
$\Pr(y \mathbf{x})$	0.2	0.5	0.3

For each model:

$$E_{y|\mathbf{x}}[(1.8 - y)^2] = (1.8 - 1)^2(0.2) + (1.8 - 2)^2(0.5) + (1.8 - 3)^2(0.3) = 0.58$$

$$E_{y|\mathbf{x}}[(2.0 - y)^2] = (2.0 - 1)^2(0.2) + (2.0 - 2)^2(0.5) + (2.0 - 3)^2(0.3) = 0.50$$

$$E_{y|\mathbf{x}}[(2.4 - y)^2] = (2.4 - 1)^2(0.2) + (2.4 - 2)^2(0.5) + (2.4 - 3)^2(0.3) = 0.58$$

Expected total test error:

$$\begin{aligned}\mathbb{E}_{S,(\mathbf{x},y)}[(h_S(\mathbf{x}) - y)^2] &= (0.3)(0.58) + (0.4)(0.50) + (0.3)(0.58) \\ &= 0.548\end{aligned}$$

Example: Bias Computation

Setup

- S is a random sample
- $\mathbf{x}$ is the input feature vector
- In this example, the input $\mathbf{X}$ takes the single value $(3, 1)$ with probability 1.
- y is the true target value for $\mathbf{x}$

Model Predictions and True Distribution

Predictions across models: Let S_1, S_2 and S_3 be three samples such that:

S	S_1	S_2	S_3
$h_S(\mathbf{x})$	1.8	2.0	2.4
$\Pr(S)$	0.3	0.4	0.3

True conditional distribution of y :

y	1	2	3
$\Pr(y \mathbf{x})$	0.2	0.5	0.3

Our goal is to compute the **squared bias**

$$\text{Bias}^2 = E_{\mathbf{x}}[(\bar{h}(\mathbf{x}) - \bar{y}(\mathbf{x}))^2] .$$

Step 1: Compute Expected Prediction and True Mean

Expected model prediction:

$$\bar{h}(\mathbf{x}) = 0.3(1.8) + 0.4(2.0) + 0.3(2.4) = 2.06$$

Expected true value:

$$\bar{y}(\mathbf{x}) = 1(0.2) + 2(0.5) + 3(0.3) = 2.1$$

$$\Rightarrow \bar{h}(\mathbf{x}) = 2.06, \quad \bar{y}(\mathbf{x}) = 2.1$$

Step 2: Computation of Bias

$$\text{Bias}^2 = (\bar{h}(\mathbf{x}) - \bar{y}(\mathbf{x}))^2$$

$$\text{Bias}^2 = (2.06 - 2.1)^2 = (-0.04)^2 = 0.0016$$

$$\boxed{\text{Bias}^2 = 0.0016}$$

Variance Definition

Definition

The variance measures the variability of model predictions across datasets:

$$\text{Variance} = E_{\mathbf{x}, S} [(h_S(\mathbf{x}) - \bar{h}(\mathbf{x}))^2]$$

Computation of Variance

In the same setup as previous, since $\mathbf{X} = (3, 1)$ with probability 1, this simplifies to:

$$\text{Variance} = E_S [(h_S(\mathbf{x}) - \bar{h}(\mathbf{x}))^2] \text{ where } \bar{h}(\mathbf{x}) = 2.06.$$

Step 1: Computation of Variance

S	$h_S(\mathbf{x})$	$h_S - \bar{h}$	$(h_S - \bar{h})^2$	$\Pr(S)$
S_1	1.8	-0.26	0.0676	0.3
S_2	2.0	-0.06	0.0036	0.4
S_3	2.4	0.34	0.1156	0.3

$$\text{Variance} = E_S[(h_S(\mathbf{x}) - \bar{h}(\mathbf{x}))^2] \text{ where } \bar{h}(\mathbf{x}) = 2.06.$$

$$\text{Variance} = 0.3(0.0676) + 0.4(0.0036) + 0.3(0.1156)$$

$$\text{Variance} = 0.02028 + 0.00144 + 0.03468 = 0.0564$$

Variance = 0.0564

Interpretation: Model predictions vary moderately across training datasets or random model states.

References

- 1 Shalev-Shwartz, Shai, and Shai Ben-David. Understanding machine learning: From theory to algorithms. Cambridge university press, 2014.
- 2 Lecture 12: Bias-Variance Tradeoff (Machine Learning for Intelligent Systems CS4780/CS5780) by Prof Kilian Weinberger, Cornell University (2018):
<https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote12.html>
- 3 Understanding the Bias-Variance Tradeoff by Scott Fortmann-Roe (2012):
<https://scott.fortmann-roe.com/docs/BiasVariance.html>
- 4 Fundamentals of Mathematical Statistics (tenth revised edition), by S.C. Gupta and V.K. Kapoor, Sultan Chand & Sons.

Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

PROF. ARINDAM BANERJEE

Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 04: Statistical Learning Theory-I

Lecture 04: Bias, Variance and Mean Squared Error



Law of Iterated Expectation for Random Vectors

Setup

Let

$$\mathbf{X} = (X_1, X_2, \dots, X_n), \quad \mathbf{Y} = (Y_1, Y_2, \dots, Y_n)$$

be random vectors defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Assume that the joint distribution of $(\mathbf{X}, \mathbf{Y})$ admits a density (or mass function) $f_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y})$ such that all expectations below are well-defined.

Statement of the Property

Law of Iterated Expectation (Vector Form)

For any integrable function $g : \mathbb{R}^{2n} \rightarrow \mathbb{R}$,

$$\mathbb{E}_{\mathbf{X}, \mathbf{Y}}[g(\mathbf{X}, \mathbf{Y})] = \mathbb{E}_{\mathbf{X}}[\mathbb{E}_{\mathbf{Y}|\mathbf{X}}[g(\mathbf{X}, \mathbf{Y})]].$$

Interpretation

- The inner expectation averages $g(\mathbf{X}, \mathbf{Y})$ over the conditional distribution of $\mathbf{Y}$ given $\mathbf{X}$.
- The outer expectation then averages over the marginal distribution of $\mathbf{X}$.

Formal Setup:

Step 1: Drawing a Data Set.

- 1 Let $S = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$ be a sample drawn independently and identically from a joint probability distribution $P(X, Y)$.
- 2 We want to find a rule $Y = h_S(X)$ in order to model Y in terms of the sample data S . Note that $y \in \mathbb{R}$.
- 3 **Step 2:** We introduce an algorithm $\mathcal{A}$ on the data set S to learn a hypothesis. Formally, we denote this by $h_S = \mathcal{A}(S)$.
- 4 **Step 3:** Realize that h_S is a random variable:
 - 1 $S = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$ is an i.i.d. sample that is drawn from a population X, Y with joint distribution $P(X, Y)$ therefore S is a random variable.
 - 2 h_S is a function of S and is therefore a random variable.

- 1 **Expected value:** For a given $\mathbf{x} \in \mathbb{R}^d$ the expected value $\bar{y}(\mathbf{x})$ is given by:

$$\bar{y}(\mathbf{x}) = E_{y|\mathbf{x}}[Y] = \int_y y \Pr(y|\mathbf{x}) dy$$

- 2 **Expected Test Error of a Given Sample:** For a given h_S learned on the dataset S , the expected test error is:

$$\mathbb{E}_{(\mathbf{x}, y) \sim P(X, Y)} [(h_S(\mathbf{x}) - y)^2] = \int_{\mathbf{x}} \int_y (h_S(\mathbf{x}) - y)^2 \Pr(\mathbf{x}, y) dy d\mathbf{x}$$

- 3 **Expected Value of Predictor:** For a given algorithm $\mathcal{A}$, the expected classifier is:

$$\bar{h} = \mathbb{E}_{S \sim P(X, Y)}[h_S] = \int_S h_S \Pr(S) dS,$$

where $\Pr(S)$ is the probability of drawing S from a population (X, Y) with joint distribution $P(X, Y)$.

Bias

The error due to bias is the difference between the expected prediction of a model and the correct value of the quantity that is being predicted.

$$\text{Bias}^2 = E_{\mathbf{x}} [(\bar{h}(\mathbf{x}) - \bar{y}(\mathbf{x}))^2]$$

Variance

The error due to variance is the variability of a model prediction for a given data point.

$$\text{Variance} = E_{\mathbf{x}, S} [(h_S(\mathbf{x}) - \bar{h}(\mathbf{x}))^2]$$

Expected Total Test Error:

For a given algorithm $\mathcal{A}$, the expected test error is:

$$\mathbb{E}_{\substack{(\mathbf{x}, y) \sim P(X, Y) \\ S \sim P(X, Y)}} [(h_S(\mathbf{x}) - y)^2] = \int_S \int_x \int_y (h_S(\mathbf{x}) - y)^2 \Pr(\mathbf{x}, y) P(S) \, d\mathbf{x} dy dS$$

where $\Pr(S)$ is the probability of drawing S from a population X, Y with joint distribution $P(X, Y)$

Example: Predicting House Prices with Elevators

Setup

True relationship:

$$Y = 200X + 100 + \varepsilon, \quad \text{where } X \sim \text{Uniform}(\{0, 1, 2\}), \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

- X : Area of a house (in 1000 ft^2)
- Y : Price of a house (in 100000 Rupees)
- Each house has an elevator that adds a fixed base cost (100 in this model).
- ε : Random effects such as location, age, or design features.

Sample

We observe a random sample:

$$S = \{(x_1, y_1), (x_2, y_2), (x_3, y_3)\},$$

where each x_i is drawn independently from $\text{Uniform}(\{0, 1, 2\})$ and

$$y_i = 200x_i + 100 + \varepsilon_i.$$

Model: A Pricing Rule

Linear Model

We fit a linear model using least squares:

$$h_S(x) = a_S x + b_S,$$

where a_S (slope) represents the estimated price per 1000 ft^2 , and b_S (intercept) represents the estimated base cost (elevator + fixed features).

Estimation

$$(a_S, b_S) = \arg \min_{a, b} \sum_{i=1}^3 (y_i - (ax_i + b))^2.$$

Since the training data vary with each sample S , both a_S and b_S are random variables.

Both a_S and b_S are unbiased estimators of the slope and intercept of the true relationship respectively (This is called Gauss- Markov Theorem).

Bias Variance in the Housing Price Model

Expected Model (Average Across Datasets)

$$\bar{h}(x) = \mathbb{E}_S[h_S(x)].$$

If the linear model is correctly specified, then:

$$\bar{h}(x) = 200x + 100,$$

and

$$\text{Bias}^2 = \mathbb{E}_X[(\bar{h}(x) - \mathbb{E}[y|x])^2] = 0.$$

Variance

The variance measures how much our predicted price changes for different training samples:

$$\text{Variance} = \mathbb{E}_X[\text{Var}_S(h_S(x))].$$

This variance increases if:

- The housing market is noisy (large σ^2),
- We have few training samples (e.g., only 3 houses),
- The training areas x_i happen to be clustered or unrepresentative.

Understanding the Bias Term

Definition

$$\text{Bias}^2 = \mathbb{E}_X [(\bar{h}(x) - \mathbb{E}[y|x])^2]$$

- Measures how far the **average model prediction** $\bar{h}(x)$ is from the **true mean response** $\mathbb{E}[y|x]$.
- Represents the systematic error of the learning algorithm.

Step 1: The Average Model Prediction

Fitted Model

We fit a linear regression model:

$$h_S(x) = a_S x + b_S,$$

where parameters (a_S, b_S) depend on the random training data S .

Expected Model (Across Many Datasets)

If we train on infinitely many random samples:

$$\bar{h}(x) = \mathbb{E}_S[h_S(x)] = 200x + 100.$$

- The average fitted line matches the true linear relationship.
- The noise averages out across datasets.

Step 2: Computing the Bias

Bias Calculation

$$\bar{h}(x) - \mathbb{E}[y|x] = (200x + 100) - (200x + 100) = 0.$$

Therefore,

$$(\bar{h}(x) - \mathbb{E}[y|x])^2 = 0 \quad \Rightarrow \quad \text{Bias}^2 = \mathbb{E}_X[0] = 0.$$

Interpretation

- On average, our model predicts the correct mean house price for every area x .
- There is **no systematic overpricing or underpricing**.

Step 3: Intuitive Understanding

Real-Life Meaning

- The model is **unbiased**: on average, it gets the pricing trend exactly right.
- Training on different samples changes the line slightly, but it oscillates around the true line.
- Bias would appear only if the model form were wrong (e.g., if true prices were nonlinear).

Understanding the Variance Term

Definition

$$\text{Variance} = \mathbb{E}_X[\text{Var}_S(h_S(x))]$$

- Measures how much the model's prediction $h_S(x)$ **changes across different training samples S** .
- Reflects how sensitive the learning algorithm is to data fluctuations.

Step 1: What Causes Model Variance?

Randomness in the Training Data

Each training sample contains:

$$S = \{(x_1, y_1), (x_2, y_2), (x_3, y_3)\}, \quad y_i = 200x_i + 100 + \varepsilon_i,$$

where x_i and ε_i are random.

Effect on the Model

- Different samples $\Rightarrow$ different fitted slopes a_S and intercepts b_S .
- Each fitted line $h_S(x) = a_S x + b_S$ is slightly different.
- Hence, for a fixed house size x , predictions $h_S(x)$ vary across datasets.

Step 3: Mathematical Expression

Variance of Predictions at a Fixed x

$$\text{Var}_S[h_S(x)] = \mathbb{E}_{X,S}[(h_S(x) - \bar{h}(x))^2] = \mathbb{E}_X[\mathbb{E}_{S \mid X=x}[(h_S(x) - \bar{h}(x))^2]] = \mathbb{E}_X[\text{Var}_{S \mid X=x}(h_S(x))]$$

- For a fixed house size, this measures how much predicted prices fluctuate across different training sets.
- Averaging this quantity over all X gives:

$$\text{Variance} = \mathbb{E}_X[\text{Var}_{S \mid X=x}(h_S(x))].$$

Step 4: What Influences the Variance

High Variance Scenario

- Very few training houses (small sample size).
- Large market noise (σ^2 is high).
- Model too flexible â fits noise instead of the true pattern.

Low Variance Scenario

- Many training houses (large sample size).
- Low market noise (prices more consistent).
- Model is stable across datasets.

Step 5: Variance in Our Linear Model

Recall the Result

In our house price model:

$$h_S(x) = a_S x + b_S, \quad \bar{h}(x) = 200x + 100.$$

Theoretical computation (from earlier derivation):

$$\text{Variance} = \mathbb{E}_{X,S}[(h_S(x) - \bar{h}(x))^2] = \mathbb{E}_X[\text{Var}_S \mid X=x(h_S(x))].$$

Using the linear model,

$$\begin{aligned} \mathbb{E}_X[\text{Var}_S \mid X=x(h_S(x))] &= \mathbb{E}_X[\text{Var}_S \mid X=x(a_S x + b)] \\ &= \mathbb{E}_X[x^2 \text{Var}_S \mid X=x(a_S) + \text{Var}_S \mid X=x(b_S) + x \text{Cov}_S \mid X=x(a_S, b_S)] \end{aligned}$$

Step 6: Variance in Our Linear Model

Further Computations

$$\mathbb{E}_X[\text{Var}_S |_{X=x}(h_S(x))] = \mathbb{E}_X[x^2 \text{Var}_S |_{X=x}(a_S) + \text{Var}_S |_{X=x}(b_S) + x \text{Cov}_S |_{X=x}(a_S, b_S)]$$

- 1 Since a_S and b_S are the least squares estimate of a and b respectively, we have closed expressions for $\mathbb{E}_X[\text{Var}_S |_{X=x}(a_S)]$, $\mathbb{E}_X[\text{Var}_S |_{X=x}(b_S)]$ and $\text{Cov}_S |_{X=x}(a_S, b_S)$
- 2 Since x_1, x_2, x_3 is drawn from a population X which takes only a finite number of values. Therefore the quantities $\mathbb{E}_X[\text{Var}_S |_{X=x}(a_S)]$, $\mathbb{E}_X[\text{Var}_S |_{X=x}(b_S)]$ and $\text{Cov}_S |_{X=x}(a_S, b_S)$ can be computed.
- 3 There are 27 possibilities for the triplet (x_1, x_2, x_3) each with probability $\frac{1}{27}$.

Note:

Note that even for a simple model such as a linear model, the computations for bias and variance can be cumbersome.

References

- 1 Shalev-Shwartz, Shai, and Shai Ben-David. Understanding machine learning: From theory to algorithms. Cambridge university press, 2014.
- 2 Lecture 12: Bias-Variance Tradeoff (Machine Learning for Intelligent Systems CS4780/CS5780) by Prof Kilian Weinberger, Cornell University (2018):
<https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote12.html>
- 3 Understanding the Bias-Variance Tradeoff by Scott Fortmann-Roe (2012):
<https://scott.fortmann-roe.com/docs/BiasVariance.html>

Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY

PROF. ARINDAM BANERJEE

Department of Mathematics
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Module 04: Statistical Learning Theory-I

Lecture 05: Bias, Variance and Mean Squared Error



Law of Iterated Expectation for Random Vectors

Setup

Let

$$\mathbf{X} = (X_1, X_2, \dots, X_n), \quad \mathbf{Y} = (Y_1, Y_2, \dots, Y_m)$$

be random vectors defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Assume that the joint distribution of $(\mathbf{X}, \mathbf{Y})$ admits a density (or mass function) $f_{\mathbf{X}, \mathbf{Y}}(\mathbf{x}, \mathbf{y})$ such that all expectations below are well-defined.

Statement of the Property

Law of Iterated Expectation (Vector Form)

For any integrable function $g : \mathbb{R}^{2n} \rightarrow \mathbb{R}$,

$$\mathbb{E}_{\mathbf{X}, \mathbf{Y}}[g(\mathbf{X}, \mathbf{Y})] = \mathbb{E}_{\mathbf{X}}[\mathbb{E}_{\mathbf{Y}|\mathbf{X}}[g(\mathbf{X}, \mathbf{Y})]].$$

Interpretation

- The inner expectation averages $g(\mathbf{X}, \mathbf{Y})$ over the conditional distribution of $\mathbf{Y}$ given $\mathbf{X}$.
- The outer expectation then averages over the marginal distribution of $\mathbf{X}$.

Formal Setup:

Step 1: Drawing a Data Set.

- 1 Let $S = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$ be a sample drawn independently and identically from a joint probability distribution $P(X, Y)$.
- 2 We want to find a rule $Y = h_S(X)$ in order to model Y in terms of the sample data S . Note that $y \in \mathbb{R}$.
- 3 **Step 2:** We introduce an algorithm $\mathcal{A}$ on the data set S to learn a hypothesis. Formally, we denote this by $h_S = \mathcal{A}(S)$.
- 4 **Step 3:** Realize that h_S is a random variable:
 - 1 $S = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$ is an i.i.d. sample that is drawn from a population X, Y with joint distribution $P(X, Y)$ therefore S is a random variable.
 - 2 h_S is a function of S and is therefore a random variable.

Bias

The error due to bias is the difference between the expected prediction of a model and the correct value of the quantity that is being predicted.

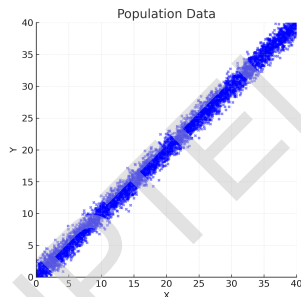
$$\text{Bias}^2 = E_{\mathbf{x}} [(\bar{h}(\mathbf{x}) - \bar{y}(\mathbf{x}))^2]$$

Variance

The error due to variance is the variability of a model prediction for a given data point.

$$\text{Variance} = E_{\mathbf{x}, S} [(h_S(\mathbf{x}) - \bar{h}(\mathbf{x}))^2]$$

Example (Setup: The True Function)

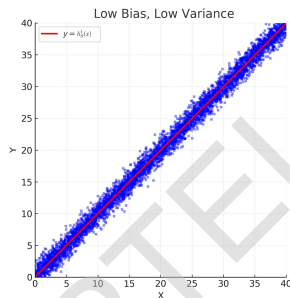


We consider the Population Data

$$Y = f(X) + \varepsilon, \quad f(x) = x, \quad X \sim \text{Uniform}[0, 40], \quad \mathbb{E}[\varepsilon] = 0, \quad \text{Var}(\varepsilon) = \sigma^2.$$

- The true expected value of Y is $\bar{y}(x) = \mathbb{E}[Y|X = x] = f(x) = x$.
- Observed data are tightly clustered around the line $Y = X$.

Definition of h_S^1



A sample S consists of $n = 5000$ points drawn from this distribution. From the training data S , we fit a straight line:

$$h_S^1(x) = a + bx,$$

where a, b are chosen to minimize the squared error on S .

Observation

Because the true relation is linear ($f(x) = x$), each fitted line will have parameters very close to

$$a \approx 0, \quad b \approx 1.$$

Hence the average fitted line aligns closely with the true function.

Bias of h_S^1

$$\text{Bias}^2 = \mathbb{E}_X [(\bar{h}_1(x) - \bar{y}(x))^2],$$

where $\bar{h}_1(x) = \mathbb{E}_S[h_S^1(x)]$.

- Since $f(x) = x$ lies in the class of linear functions,

$$\mathbb{E}_S[h_S^1(x)] = x = \bar{y}(x).$$

- Therefore,

$$\boxed{\text{Bias}^2 = 0.}$$

Interpretation

The model is not systematically off, it does capture the true pattern exactly.

Variance of h_S^1

$$\text{Variance} = \mathbb{E}_{X,S} [(h_S^1(x) - \bar{h}_1(x))^2].$$

Different random samples S produce slightly different (a, b) values. However, with many points spread over $[0, 40]$, this variation becomes very small.

$$\text{Variance} = O\left(\frac{\sigma^2}{n}\right).$$

For our case

With $n = 5000$,

Variance is extremely low.

Conclusion

From our definitions:

$$\text{Bias}^2 = \mathbb{E}_X[(\bar{h}_1(x) - \bar{y}(x))^2] = 0,$$

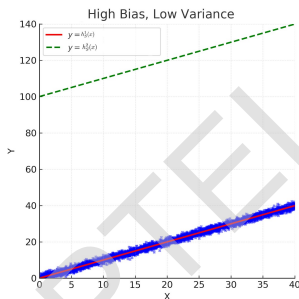
$$\text{Variance} = \mathbb{E}_{X,S}[(h_S^1(x) - \bar{h}(x))^2] \text{ is small.}$$

Conclusion

For $X \sim \text{Uniform}[0, 40]$ and $Y = X + \varepsilon$, the fitted line $h_S^1(x) = a + bx$ has

h_S^1 has **Low Bias** and **Low Variance**.

Definition of h_S^2



Definition

As before, let:

$$h_S^1(x) = ax + b$$

and define a new shifted predictor

$$h_S^2(x) = h_S^1(x) + 100.$$

Here, the shift by +100 represents a systematic offset.

Expectation of h_S^2

Expected Predictor

For the original model family $\{h_S^1(x)\}$:

$$\bar{h}_1(x) = \mathbb{E}_S[h_S^1(x)]$$

For the shifted version:

$$\begin{aligned}\bar{h}_2(x) &= \mathbb{E}_S[h_S^2(x)] \\ &= \mathbb{E}_S[h_S^1(x) + 100] \\ &= \bar{h}_1(x) + 100.\end{aligned}$$

Observation

Shifting by a constant moves the expected prediction by the same constant, but does not change its variability across samples.

Bias and Variance of h_S^2

Definitions (from earlier)

$$\text{Bias}^2 = \mathbb{E}_X[(\bar{h}_2(x) - \bar{y}(x))^2], \quad \text{Variance} = \mathbb{E}_{X,S}[(h_S^2(x) - \bar{h}_2(x))^2].$$

Computation for the shifted model

$$\text{Bias}_2^2 = \mathbb{E}_X[(\bar{h}_2(x) - \bar{y}(x))^2] = \mathbb{E}_X[(\bar{h}_1(x) - \bar{y}(x) + 100)^2] \approx 100^2 = 10,000,$$

$$\text{Variance}_2 = \mathbb{E}_{X,S}[(h_S^2(x) - \bar{h}_2(x))^2] = \mathbb{E}_{X,S}[(h_S^1(x) - \bar{h}_1(x))^2] = \text{Variance}_1.$$

Conclusion

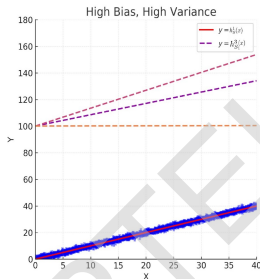
Key Points

- Adding a constant +100 **does not change** the spread of model predictions across different samples $\Rightarrow$ variance remains small.
- It introduces a **large systematic offset**, making every prediction 100 units too high.

Conclusion

$h_S^2(x)$ has **High Bias** and **Low Variance**.

Definition of h_S^3



Shifted and Random Slope Predictor

Let $S = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ be the sample drawn.

$$h_S^3(x) = h_S^1(x) + 100 + (x_1 - 20)x. \quad \text{where } h_S^1(x) \text{ is the best fit line using } \{(x_2, y_2), \dots, (x_n, y_n)\}$$

Expectation over Samples

$$\overline{h_3}(x) = \mathbb{E}_S[h_S^3(x)] = \overline{h_1}(x) + 100 + \mathbb{E}_S[(x_1 - 20)]x = \overline{h_1}(x) + 100 \text{ since } \mathbb{E}_S[(x_1 - 20)] = 0.$$

Bias of h_S^3

Note

Note that $\bar{h}_3(x) = \bar{h}_1(x) - 100$ as the sample is i.i.d.

Definition

$$\text{Bias}^2 = \mathbb{E}_X [(\bar{h}_3(x) - \bar{y}(x))^2]$$

Computation

The model h_S^1 is unbiased since it is a least squares line, $\bar{h}_1(x) = \bar{y}(x) = f(x)$, then

$$\text{Bias}_3^2 = \mathbb{E}_x [(f(x) + 100 - f(x))^2] = \mathbb{E}_x [100^2] = \boxed{10,000.}$$

Hence the constant +100 produces a large systematic error $\Rightarrow$ **high bias**.

Variance of h_S^3

Definition

$$\text{Variance} = \mathbb{E}_{X,S} [(h_S^3(x) - \bar{h}_3(x))^2] .$$

Computation for h_S^3

$$\begin{aligned} \text{Variance}_3 &= \mathbb{E}_{X,S} [(h_S^3(x) - \bar{h}_3(x))^2] \\ &= \mathbb{E}_{X,S} [(h_S^1(x) + (x_1 - 20)x - \bar{h}_1(x))^2] \\ &= \mathbb{E}_{X,S} [(h_S^1(x) - \bar{h}_1(x))^2 + ((x_1 - 20)x)^2 + 2(h_S^1(x) - \bar{h}_1(x))(x_1 - 20)x] \\ &= \mathbb{E}_{X,S} [(h_S^1(x) - \bar{h}_1(x))^2] + \mathbb{E}_{X,S} [((x_1 - 20)x)^2] \end{aligned}$$

The cross terms vanish by the fact that

$$\mathbb{E}_{X,S} [2(h_S^1(x) - \bar{h}_1(x))(x_1 - 20)x] = \mathbb{E}_{S|X=x} [2(h_S^1(x) - \bar{h}_1(x))(x_1 - 20)x] = \mathbb{E}_{S|X=x} [2(h_S^1(x) - \bar{h}_1(x))] \mathbb{E}_{S|X=x} [(x_1 - 20)x] = 0$$

since we have $\mathbb{E}[x_1 - 20] = 0$.

Result (for $X \sim \text{Uniform}[0, 40]$)

$$\text{Variance}_3 = \text{Variance}_1 + \text{Var}(x_1 - 20)\mathbb{E}[x^2], \quad \mathbb{E}[x^2] = \frac{40^2}{3} = \frac{1600}{3} \approx 533.33.$$

$$\boxed{\text{Variance}_3 \approx \text{Variance}_1 + 133.33 \times 533.33}$$

Conclusion

Summary

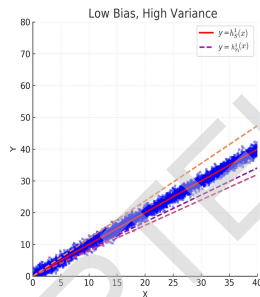
$$\begin{aligned}\text{Bias}_3^2 &= 10,000, \\ \text{Variance}_3 &= \text{Variance}_1 + 133.33 \times 533.33\end{aligned}$$

Interpretation

- The constant shift $+100 \Rightarrow$ very **high bias**.
- The random slope term $B_S x \Rightarrow$ very **high variance**.

$h_S^3(x)$ has **High Bias** and **High Variance**.

Definition of $h_S^4(x)$



New Model

Let $S = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ be the sample drawn.

$$h_S^4(x) = h_S^1(x) + (x_1 - 20)x. \text{ where } h_S^1(x) \text{ is the best fit line using } \{(x_2, y_2), \dots, (x_n, y_n)\}$$

Interpretation

Each dataset S produces a slightly different slope due to $(x_1 - 20)$, so h_S^4 fluctuates around h_S^1 .

Bias and Variance of h_S^4

Bias

$$\mathbb{E}_S[h_S^4(x)] = \mathbb{E}_S[h_S^1(x) + (x_1 - 20)x] = h_S^1(x)$$

Hence, h_S^4 is centered near the true model $\Rightarrow$ ****Low Bias.****

Variance

$$\text{Var}_S[h_S^4(x)] = \mathbb{E}_S[((x_1 - 20)x)^2] = 133.33x^2$$

Variance grows quickly with $x \Rightarrow$ High Variance.

Conclusion

Conclusion

$h_S^4(x)$ has **Low Bias** (correct mean) but **High Variance** (spread of slopes across samples)

References

- 1 Shalev-Shwartz, Shai, and Shai Ben-David. Understanding machine learning: From theory to algorithms. Cambridge university press, 2014.
- 2 Lecture 12: Bias-Variance Tradeoff (Machine Learning for Intelligent Systems CS4780/CS5780) by Prof Kilian Weinberger, Cornell University (2018):
<https://www.cs.cornell.edu/courses/cs4780/2018fa/lectures/lecturenote12.html>
- 3 Understanding the Bias-Variance Tradeoff by Scott Fortmann-Roe (2012):
<https://scott.fortmann-roe.com/docs/BiasVariance.html>

Thank you

